

Source Exclusion in Regime-Gated Forecasting: A Cross-Market Audit of Equity-Factor Predictability

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Removing lag coefficients can leave the source inside the regime gate.

RESEARCH QUESTION

How does measured predictive value change when a factor is removed from explicit regression lags versus every input path of a regime-gated forecasting procedure?

42 / 240

unweighted temporal sign disagreements

0 / 2,400

directional interval crossings

776

margin inclusions explained by proximity

1 • Three distinct questions

Shared-gate ablation asks whether explicit source lags add value given a source-informed gate. Complete exclusion removes the source from fitting, filtering, scaling, and tuning. Gate-value comparison tests gated versus pooled forecasts with the same raw information. These are different estimands; shared-gate ablation is not inherently invalid.

3 • Empirical removal sensitivity

Gain is restricted-procedure loss minus source-included loss; positive favors inclusion. Complete exclusion and shared-gate ablation have opposite capped-loss signs in 42 of 240 unweighted temporal HMM comparisons. Both component gains lie within ± 0.005 ; the median absolute direct contrast is 0.000341. These are descriptive discrepancies, not resolved population reversals.

5 • Known-information controls

Constructed simulations separate oracle information value from fitted procedure costs. At 10% binary-source corruption, oracle information gain is +0.0708 while Gaussian- and Student-t-HMM gains are -0.0494 and -0.0486. Positive information can coexist with harmful fitted inclusion.

2 • Complete family and chronology

Six equity factors, all 30 ordered directions, and four regions: US, Europe, Japan, and Asia Pacific excluding Japan. Initial training precedes 2005, validation covers 2005-2009, and evaluation uses annual refits. January 2025-August 2026 is reported separately from inspected history.

4 • Conditional inference

The normalized score is $\min(\text{squared forecast error} / c, 1)$, with c fixed from training. Across 2,400 temporal endpoints, none has a simultaneous directional crossing. All 776 intervals inside ± 0.005 are explained by forecast proximity; 196 concern identical forecasts. The margin measures score resolution, not economic utility.

EVIDENCE AT A GLANCE

Source access	Full model	Shared-gate ablation	Complete exclusion
Explicit source lags	Yes	No	No
Gate fit and filter	Yes	Yes	No
Restricted preprocessing/tuning	Source-informed	Source-informed	Source-excluded

TAKEAWAY

Source-removal definitions affect measured gains. The audit finds limited directional resolution and shows why fitted losses cannot establish either universal predictive success or absence of oracle information.

INTERPRETATION & LIMITS

Inference targets normalized capped squared loss, not ordinary MSE or investment utility. Regions and directions are dependent; data vintages are retrospective. Later evaluation was outside prior repository analysis, but outside exposure is unverified. Simulations do not identify a financial mechanism.